Model Parameter Inventory

Hard-Coded Input Parameters for Model Governance

MKM Research Labs

Version 1.0 — 16-February-2026

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Contents

1 Executive Summary

This document provides a complete inventory of all hard-coded input parameters used by the MKM Physical Risk analytical models. It is maintained as part of the Model Risk Governance framework and should be reviewed alongside each model's documentation.

1.1 Purpose

Hard-coded parameters include physical constants, default function arguments, calibration values, threshold tables, and factor look-ups. Documenting these centrally enables:

- Audit trail for model validation and MRC review
- Impact assessment when parameters are recalibrated
- Identification of cross-model dependencies on shared constants

1.2 Scope

This inventory covers **13** model modules containing the parameters listed below. Each entry provides the parameter name, current value, description, and source file with line number for traceability.

Generated on **16 February 2026**.

2 PRS Pricing — Analytical Model (MKM-PR-001)

Source: `models/hazard/prs_analytical.py`

2.1 Basis Waterfall Components

Parameter	Value	Description	Source
`MODEL_UNCERTAINTY_BPS`	2.0	Fixed model uncertainty spread (basis points)	`hazard/prs_analytical.py:20`
`DISTANCE_RATE_BPS_PER_KM`	0.5	Distance basis rate per km	`hazard/prs_analytical.py:46`
`DISTANCE_MAX_BPS`	6.0	Distance basis cap (bps)	`hazard/prs_analytical.py:47`
`DISTANCE_CAP_KM`	12.0	Maximum distance for basis	`hazard/prs_analytical.py:48`
`ELEVATION_RATE_BPS_PER_M`	0.2	Elevation benefit rate per metre	`hazard/prs_analytical.py:51`
`ELEVATION_MAX_BENEFIT_BPS`	3.0	Maximum elevation benefit cap (bps)	`hazard/prs_analytical.py:52`
`MIN_PRS_SPREAD_BPS`	2.0	Minimum PRS spread floor (bps)	`hazard/prs_analytical.py:62`
`CONSTRUCTION_YEAR_CUTOFF`	2000	Pre-cutoff adds 1.0 bps penalty	`hazard/prs_analytical.py:43`

2.2 Terrain Basis by EA Flood Zone (bps)

Parameter	Value	Description	Source
`Zone 3b / 3a / 3`	3.0	High flood risk zones	`hazard/prs_analytical.py:23-`
`Zone 2`	2.0	Medium flood risk zone	`hazard/prs_analytical.py:23-`
`Zone 1`	0.0	Low flood risk zone	`hazard/prs_analytical.py:23-`
`Enhanced Defence`	-1.0	Protected area benefit	`hazard/prs_analytical.py:23-`

2.3 Composition Basis by Property Type (bps)

Parameter	Value	Description	Source
`Flat`	2.0	Higher risk construction type	`hazard/prs_analytical.py:33-`
`Bungalow`	1.0	Ground-floor exposure	`hazard/prs_analytical.py:33-`
`Terraces`	0.5	Moderate exposure	`hazard/prs_analytical.py:33-`
`Semi-detached / Detached`	0.0	Base case	`hazard/prs_analytical.py:33-`

2.4 Recovery Rates by Trigger

Parameter	Value	Description	Source
any_flood (> 0m)	0.85	Recovery rate for any flood trigger	hazard/prs_analytical.py:55-
moderate (> 0.5m)	0.70	Recovery rate for moderate flood	hazard/prs_analytical.py:55-
severe (> 1.0m)	0.50	Recovery rate for severe flood	hazard/prs_analytical.py:55-

2.5 CDS Pricing

Parameter	Value	Description	Source
dt	0.25	Quarterly period length (years)	hazard/prs_analytical.py:97
max_annual_hazard_rate	0.999	Cap for continuous hazard rate	hazard/prs_analytical.py:95

3 PRS Pricing — QuantLib CDS (MKM-PR-001)

Source: `models/prs_with_hazard_curve.py`

3.1 Default Trade Parameters

Parameter	Value	Description	Source
`notional`	10,000,000	Default notional (GBP)	`prs_with_hazard_curve.py:131`
`tenor_years`	5	Default swap tenor	`prs_with_hazard_curve.py:132`
`running_spread`	0.01	Default running spread (100 bps)	`prs_with_hazard_curve.py:133`
`recovery_rate`	0.0	Default recovery (no recovery for flood)	`prs_with_hazard_curve.py:134`
`risk_free_rate`	0.03	Default risk-free rate (3\{}%)	`prs_with_hazard_curve.py:135`
`implied_hazard_rate fallback`	0.05	Fallback hazard rate (5\{}%)	`prs_with_hazard_curve.py:83`

3.2 QuantLib Conventions

Parameter	Value	Description	Source
`Schedule frequency`	Quarterly	CDS payment frequency	`prs_with_hazard_curve.py:194`
`Calendar`	TARGET()	TARGET settlement calendar	`prs_with_hazard_curve.py:182`
`Day counter`	Actual/360	Accrual day count convention	`prs_with_hazard_curve.py:184`
`Business day convention`	Following	Roll convention	`prs_with_hazard_curve.py:183`
`Date generation rule`	TwentiethIMM	IMM date schedule	`prs_with_hazard_curve.py:198`

4 GEV Hazard Curve Model (MKM-GH-001)

Source: `models/hazard/gev.py`, `models/hazard/builder.py`

4.1 Hazard Curve Construction

Parameter	Value	Description	Source
`STANDARD_THRESHOLDS`	[0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0]	Flood depth thresholds (m)	`hazard/builder.py:24`
`RETURN_PERIODS`	[2, 5, 10, 20, 50, 100]	Standard return periods (years)	`hazard/builder.py:25`
`exceedance_prob floor`	0.01	Floor at 1\{}% (100-yr return period)	`hazard/builder.py:69-70`

4.2 Term Structure

Parameter	Value	Description	Source
`max_years`	5	Maximum years for term structure	`hazard/gev.py:48`

5 Storm-Gauge Response Model (MKM-SG-001)

Source: `models/hazard/response_model.py`, `models/stormgauge/forward_model.py`

5.1 Gauge Alert Thresholds (defaults)

Parameter	Value	Description	Source
`flood_alert`	3.0	Flood alert level (m)	`hazard/response_model.py:42`
`flood_warning`	4.0	Flood warning level (m)	`hazard/response_model.py:43`
`severe_flood_warning`	5.0	Severe warning level (m)	`hazard/response_model.py:44`

5.2 Response Calibration

Parameter	Value	Description	Source
`base_response`	level_range * 0.015	Base gauge response coefficient	`hazard/response_model.py:59`
`response_coef variation`	U(0.8, 1.2)	Uniform multiplier on response	`hazard/response_model.py:60`
`precip_factor divisor`	35.0	Precipitation normalisation (mm)	`hazard/response_model.py:78`
`duration_factor cap`	2.0	Maximum duration amplification	`hazard/response_model.py:79`
`noise factor`	U(0.85, 1.15)	Random noise multiplier	`hazard/response_model.py:89`

5.3 Forward Model — Spatial Decay

Parameter	Value	Description	Source
`intensity_to_level_scale`	0.1	Intensity-to-level conversion scale	`stormgauge/forward_model.py`
`time_resolution_hours`	0.5	Temporal resolution (hours)	`stormgauge/forward_model.py`
`response_lag_hours`	2.0	Lag before gauge response	`stormgauge/forward_model.py`
`response_decay_hours`	12.0	Exponential decay time	`stormgauge/forward_model.py`
`gamma (intensity mapping)`	0.8	Non-linear intensity exponent	`stormgauge/forward_model.py`
`rise limb multiplier`	3.0	Faster rising limb vs decay	`stormgauge/forward_model.py`
`Earth radius`	6,371 km	Haversine distance calculation	`stormgauge/forward_model.py`

6 Storm Intensity Distribution (MKM-SI-001)

Source: `models/intensity/distribution.py`, `models/intensity/parameters.py`

6.1 Distribution Defaults

Parameter	Value	Description	Source
`base_mean`	45.0	Mean storm intensity	`intensity/distribution.py:3`
`base_std`	15.0	Standard deviation	`intensity/distribution.py:3`
`tail_threshold`	60.0	Pareto tail crossover point	`intensity/distribution.py:3`
`tail_index`	2.5	Pareto tail index (heavier = lower)	`intensity/distribution.py:3`
`min_intensity`	10.0	Minimum storm intensity	`intensity/distribution.py:3`
`max_intensity`	100.0	Maximum storm intensity	`intensity/distribution.py:3`

6.2 Scenario Families

Parameter	Value	Description	Source
`historical`	mean=40, std=12, thresh=60, idx=3.0	Historical baseline scenario	`intensity/parameters.py:43-`
`baseline`	mean=45, std=15, thresh=60, idx=2.5	Current climate scenario	`intensity/parameters.py:51-`
`moderate_stress`	mean=50, std=15, thresh=55, idx=2.0	Moderate climate stress	`intensity/parameters.py:59-`
`severe_stress`	mean=55, std=18, thresh=50, idx=1.5	Severe climate stress	`intensity/parameters.py:67-`
`extreme`	mean=60, std=20, thresh=45, idx=1.2	Extreme tail scenario	`intensity/parameters.py:75-`

7 Flood Risk — Depth-Damage and Velocity (MKM-DD-001)

Source: `models/floodrisk/depth_damage.py`, `models/floodrisk/velocity.py`

7.1 UK Depth-Damage Curve

Parameter	Value	Description	Source
`DEPTH_POINTS`	[0, 0.05, 0.5, 1, 1.5, 2, 3, 4, 5, 6]	Flood depth breakpoints (m)	`floodrisk/depth_damage.py:80`
`DAMAGE_POINTS`	[0, 0.05, 0.25, 0.4, 0.5, 0.6, 0.75, 0.85, 0.95, 1.0]	Damage fraction at each depth	`floodrisk/depth_damage.py:81`
`max flood depth`	5.0	Maximum modelled depth (m)	`floodrisk/depth_damage.py:74`
`max distance`	25,000	Maximum distance for damage (m)	`floodrisk/depth_damage.py:67`

7.2 Property Type Factors

Parameter	Value	Description	Source
`residential`	1.0	Base case damage factor	`floodrisk/depth_damage.py:13`
`commercial`	1.2	Higher damage for commercial	`floodrisk/depth_damage.py:13`
`industrial`	0.9	Lower damage for industrial	`floodrisk/depth_damage.py:13`

7.3 Manning Velocity Model

Parameter	Value	Description	Source
`DEFAULT_ROUGHNESS`	0.04	Manning n for urban floodplain	`floodrisk/velocity.py:15`
`MIN_SLOPE`	0.001	Minimum slope clamp	`floodrisk/velocity.py:21`
`DEFAULT_ATTENUATION_LENGTH`	2,000	Exponential decay length (m)	`floodrisk/velocity.py:18`
`DEFAULT_RECESSION_FACTOR`	1.5	Recession limb multiplier	`floodrisk/velocity.py:24`

8 Spatial Interpolation (MKM-SP-001)

Source: `models/floodrisk/spatial.py`

8.1 Spatial Parameters

Parameter	Value	Description	Source
`Earth radius`	6,371,000	Haversine radius (m)	`floodrisk/spatial.py:53`
`IDW power`	2.0	Inverse distance weighting exponent	`floodrisk/spatial.py:114`
`IDW min_distance`	1.0	Minimum distance floor (m)	`floodrisk/spatial.py:115`
`distance_km to degrees`	111.0	km per degree latitude	`floodrisk/spatial.py:40`

9 Risk Analytics (MKM-RA-001)

Source: `models/floodrisk/risk_analytics.py`

9.1 Monte Carlo Settings

Parameter	Value	Description	Source
`n_simulations`	1,000	Default simulation count	`floodrisk/risk_analytics.py`
`shock_multiplier`	0.2	Stress shock (20\{}%)	`floodrisk/risk_analytics.py`
`random_seed`	42	Reproducibility seed	`floodrisk/risk_analytics.py`
`grid_size`	1,000	Spatial grid cell size (m)	`floodrisk/risk_analytics.py`

9.2 Risk Classification Thresholds

Parameter	Value	Description	Source
`high_risk`	¿ 0.6	Damage ratio for high risk	`floodrisk/risk_analytics.py`
`medium_risk`	0.3–0.6	Damage ratio for medium risk	`floodrisk/risk_analytics.py`
`low_risk`	0.1–0.3	Damage ratio for low risk	`floodrisk/risk_analytics.py`

10 Risk Assessment Model (MKM-RA-001)

Source: `models/risk/risk_assessor.py`

10.1 Flood Depth Thresholds

Parameter	Value	Description	Source
`very_low`	0.0 m	No flood exposure	`risk/risk_assessor.py:23-29`
`low`	0.1 m	Minor flooding	`risk/risk_assessor.py:23-29`
`medium`	0.5 m	Moderate flooding	`risk/risk_assessor.py:23-29`
`high`	1.0 m	Significant flooding	`risk/risk_assessor.py:23-29`
`very_high`	2.0 m	Severe flooding	`risk/risk_assessor.py:23-29`

10.2 LTV Risk Thresholds

Parameter	Value	Description	Source
`low`	0.6	Low LTV boundary	`risk/risk_assessor.py:31-36`
`moderate`	0.8	Moderate LTV boundary	`risk/risk_assessor.py:31-36`
`high`	0.95	High LTV boundary	`risk/risk_assessor.py:31-36`
`critical`	1.0	Negative equity threshold	`risk/risk_assessor.py:31-36`

10.3 Flood Risk Scores

Parameter	Value	Description	Source
`Very Low`	1	Base risk score	`risk/risk_assessor.py:154-1`
`Low`	2		`risk/risk_assessor.py:154-1`
`Medium`	4		`risk/risk_assessor.py:154-1`
`High`	7		`risk/risk_assessor.py:154-1`
`Very High`	9		`risk/risk_assessor.py:154-1`
`Unknown`	3	Default when unknown	`risk/risk_assessor.py:154-1`

10.4 LTV Multipliers

Parameter	Value	Description	Source
`> 0.95`	2.0x	Near negative equity	`risk/risk_assessor.py:171-1`
`> 0.8`	1.5x	High LTV	`risk/risk_assessor.py:171-1`
`> 0.6`	1.2x	Moderate LTV	`risk/risk_assessor.py:171-1`
`else`	1.0x	Low LTV (base)	`risk/risk_assessor.py:171-1`

10.5 Construction Type Factors

Parameter	Value	Description	Source
`timber / wood`	1.3	Highest vulnerability	`risk/risk_assessor.py:189-1`
`brick`	1.0	Standard	`risk/risk_assessor.py:189-1`
`concrete`	0.9	More resilient	`risk/risk_assessor.py:189-1`
`steel`	0.8	Most resilient	`risk/risk_assessor.py:189-1`

10.6 Age Factors

Parameter	Value	Description	Source
`> 100 years`	1.3	Heritage property risk	`risk/risk_assessor.py:181-1`
`> 50 years`	1.1	Older property risk	`risk/risk_assessor.py:181-1`

10.7 Insurance Premium Base Rates

Parameter	Value	Description	Source
`Very Low`	0.001	0.1\{}% of property value	`risk/risk_assessor.py:391-3`
`Low`	0.002	0.2\{}%	`risk/risk_assessor.py:391-3`
`Medium`	0.005	0.5\{}%	`risk/risk_assessor.py:391-3`
`High`	0.015	1.5\{}%	`risk/risk_assessor.py:391-3`
`Very High`	0.035	3.5\{}%	`risk/risk_assessor.py:391-3`
`Premium cap`	0.10	Max 10\{}% of property value	`risk/risk_assessor.py:406`

10.8 Distance Risk Factors

Parameter	Value	Description	Source
`<= 0.1 km`	1.0	Immediate proximity	`risk/risk_assessor.py:361-3`
`<= 0.5 km`	0.8		`risk/risk_assessor.py:361-3`
`<= 1.0 km`	0.6		`risk/risk_assessor.py:361-3`
`<= 2.0 km`	0.4		`risk/risk_assessor.py:361-3`
`<= 5.0 km`	0.2		`risk/risk_assessor.py:361-3`
`> 5.0 km`	0.1	Minimal proximity risk	`risk/risk_assessor.py:361-3`

11 Property Valuation Model (MKM-PV-001)

Source: `models/valuation/property_value.py`

11.1 Floor Area Ranges (sq m)

Parameter	Value	Description	Source
`Flat`	35–120	Typical flat range	`valuation/property_value.py`
`Mid-terrace`	60–180		`valuation/property_value.py`
`End-terrace`	70–200		`valuation/property_value.py`
`Semi-detached`	80–250		`valuation/property_value.py`
`Detached`	120–400		`valuation/property_value.py`
`Bungalow`	80–200		`valuation/property_value.py`

11.2 Base Price per sqm (GBP)

Parameter	Value	Description	Source
`Detached`	8,000–15,000	London market calibration	`valuation/property_value.py`
`Semi-detached`	6,500–12,000		`valuation/property_value.py`
`Mid-terrace`	6,000–11,000		`valuation/property_value.py`
`End-terrace`	6,200–11,500		`valuation/property_value.py`
`Bungalow`	7,000–13,000		`valuation/property_value.py`
`Flat`	5,500–10,000		`valuation/property_value.py`

11.3 Age Band Factors (multiplier range)

Parameter	Value	Description	Source
`New build (< 10y)`	1.05–1.15	Premium for new	`valuation/property_value.py`
`Modern (10--24y)`	0.95–1.05	Near base	`valuation/property_value.py`
`Established (25--49y)`	0.90–1.00	Minor discount	`valuation/property_value.py`
`Period (50--99y)`	0.85–0.95		`valuation/property_value.py`
`Heritage (100+y)`	0.80–1.20	Wide range (character premium)	`valuation/property_value.py`

11.4 Condition Factors

Parameter	Value	Description	Source
`Excellent`	1.10–1.20		`valuation/property_value.py`

Parameter	Value	Description	Source
`Good`	1.00–1.10		`valuation/property_value.py`
`Fair`	0.90–1.00		`valuation/property_value.py`
`Poor`	0.70–0.90		`valuation/property_value.py`
`Very poor`	0.50–0.70		`valuation/property_value.py`

11.5 Flood Risk Factors

Parameter	Value	Description	Source
`Very Low`	1.00–1.02	Near neutral impact	`valuation/property_value.py`
`Low`	0.98–1.00		`valuation/property_value.py`
`Medium`	0.92–0.98		`valuation/property_value.py`
`High`	0.85–0.92		`valuation/property_value.py`
`Very High`	0.75–0.85	Significant discount	`valuation/property_value.py`

11.6 EPC Rating Factors

Parameter	Value	Description	Source
`A`	1.05–1.10	Green premium	`valuation/property_value.py`
`B`	1.02–1.05		`valuation/property_value.py`
`C`	1.00–1.02		`valuation/property_value.py`
`D`	0.98–1.00		`valuation/property_value.py`
`E`	0.95–0.98		`valuation/property_value.py`
`F`	0.90–0.95		`valuation/property_value.py`
`G`	0.85–0.90	Energy penalty	`valuation/property_value.py`

11.7 Valuation Bounds

Parameter	Value	Description	Source
`MIN_PROPERTY_VALUE`	150,000	Floor (GBP)	`valuation/property_value.py`
`MAX_PROPERTY_VALUE`	5,000,000	Cap (GBP)	`valuation/property_value.py`

12 Insurance Premium Model (MKM-IP-001)

Source: `models/valuation/insurance.py`

12.1 Property Type Premium Factors

Parameter	Value	Description	Source
`Flat`	0.80–1.00	Lower exposure	`valuation/insurance.py:20-2`
`Mid-terrace`	0.90–1.10		`valuation/insurance.py:20-2`
`End-terrace`	1.00–1.20		`valuation/insurance.py:20-2`
`Semi-detached`	1.10–1.30		`valuation/insurance.py:20-2`
`Detached`	1.20–1.50	Highest exposure	`valuation/insurance.py:20-2`
`Bungalow`	1.00–1.20	Ground-floor risk	`valuation/insurance.py:20-2`

12.2 Flood Risk Premium Factors

Parameter	Value	Description	Source
`Very Low`	0.90–1.00		`valuation/insurance.py:30-3`
`Low`	1.00–1.20		`valuation/insurance.py:30-3`
`Medium`	1.30–1.80		`valuation/insurance.py:30-3`
`High`	1.80–2.50		`valuation/insurance.py:30-3`
`Very High`	2.50–4.00	Significant loading	`valuation/insurance.py:30-3`

12.3 Premium Bounds

Parameter	Value	Description	Source
`BASE_RATE_RANGE`	1.5–4.0	Per GBP 1,000 insured	`valuation/insurance.py:64`
`MIN_PREMIUM`	200	Floor (GBP)	`valuation/insurance.py:67`
`MAX_PREMIUM`	20,000	Cap (GBP)	`valuation/insurance.py:68`

13 Mortgage Pricer (MKM-MP-001)

Source: `models/mortgage_pricer.py`

13.1 Credit Spread Schedule

Parameter	Value	Description	Source
`Affordability ratio points`	[0.1, 0.2, ..., 1.0]	10 interpolation points	`mortgage_pricer.py:55`
`Credit spreads`	[50, 100, 200, 300, 500, 800, 1200, 1800, 2500, 3500] bps	Corresponding spread curve	`mortgage_pricer.py:58`
`Default for zero income`	0.15	15\{}% spread fallback	`mortgage_pricer.py:141`
`Minimum spread`	0.001	10 bps floor	`mortgage_pricer.py:175`

13.2 Flood Risk Multipliers

Parameter	Value	Description	Source
`Very Low`	1.00	No adjustment	`mortgage_pricer.py:178-184`
`Low`	1.05		`mortgage_pricer.py:178-184`
`Medium`	1.20		`mortgage_pricer.py:178-184`
`High`	1.40		`mortgage_pricer.py:178-184`
`Very High`	1.75	Largest loading	`mortgage_pricer.py:178-184`

13.3 LTV Impact Thresholds

Parameter	Value	Description	Source
`> 0.95`	1.5x	Near negative equity	`mortgage_pricer.py:389-396`
`> 0.9`	1.3x		`mortgage_pricer.py:389-396`
`> 0.8`	1.1x		`mortgage_pricer.py:389-396`
`else`	1.0x	Low LTV (base)	`mortgage_pricer.py:389-396`

13.4 Default Assumptions

Parameter	Value	Description	Source
`tax_rate`	0.20	Corporation tax (20\{}%)	`mortgage_pricer.py:37`

Parameter	Value	Description	Source
`gross_annual_income` (batch)	50,000	Batch pricing default	`mortgage_pricer.py:415`
`interest_rate` (batch)	0.035	3.5\{}% default rate	`mortgage_pricer.py:417`
`insurance_rate` (batch)	0.002	0.2\{}% insurance	`mortgage_pricer.py:418`
`recovery_haircut` (batch)	0.2	20\{}% recovery haircut	`mortgage_pricer.py:420`

14 Timeseries Statistics (MKM-TS-001)

Source: `models/statistics/timeseries.py`, `models/statistics/synthetic.py`

14.1 Statistical Parameters

Parameter	Value	Description	Source
`days_per_year`	365.25	Year length for annual maxima	`statistics/timeseries.py:48`
`percentile_bands`	[5, 10, 25, 50, 75, 90, 95, 99]	Standard percentile breakpoints	`statistics/timeseries.py:69`

14.2 Synthetic Water Level Generation

Parameter	Value	Description	Source
`base_level range`	U(0.3, 0.5) * flood_alert	Normal water level	`statistics/synthetic.py:62`
`daily_std`	base_level * 0.15	Daily standard deviation	`statistics/synthetic.py:63`
`seasonal_amplitude`	base_level * 0.25	Winter/summer variation	`statistics/synthetic.py:64`
`seasonal_peak`	Day 15 (January)	Winter peak offset	`statistics/synthetic.py:98`
`minor_flood_prob`	5.0x severe_prob	Minor flood frequency	`statistics/synthetic.py:83`

15 Document History

Date	Version	Description	Author
16-Feb-2026	1.0	Initial parameter inventory	D. Kelly